

SOLUTION ARCHITECTURE

Team ID: SWTID-2026-6530

Project Name: Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Maximum Marks: 5 Marks

Solution Architecture Diagram

Presentation / Client Layer

ServiceNow Service Catalog

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API Gateway / Load Balancer

ServiceNow Platform / ServiceNow Request Processing

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Auth Service

ServiceNow User Authentication & Access Control

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Core Logic Service

ServiceNow Flow Designer

- Service Catalog Trigger
- Approval Validation
- Create Catalog Task
- Update Short Description
- Assign Task to Hardware Group

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External API / Integrations

ServiceNow Service Catalog & Hardware Team Assignment

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Data / Storage Layer

ServiceNow Database

- Service Request
- Requested Item
- Catalog Task
- Approval Details
- Assignment Information

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Allows users to place standard laptop requests and view request/order status.	ServiceNow Service Catalog
API Gateway	Handles and processes service requests within the ServiceNow platform.	ServiceNow Platform
Auth Service	Provides authenticated access to ServiceNow and ensures users can access the required services and records.	ServiceNow Authentication & Access Control
Core Logic Service	Automates the laptop procurement workflow, checks approval, creates Catalog Tasks and assigns them to the Hardware group.	ServiceNow Flow Designer
External Integration	Connects the approved request with the Hardware team for laptop configuration activities.	ServiceNow Assignment Group
Database	Stores service requests, requested items, approvals, catalog tasks and assignment details.	ServiceNow Database